

CHANGING.AI: Delivering deep-learning- powered personalized recommendati ons services to retailers

Tell us your challenge. We're here to help.

Contact us (/contact)

CHANGING.AI

CHANGING.AI uses
Cloud to train deep
networks that can
complex calculations
across large e-commerce
providers in less than
day. This helps providers
deliver real-time,
personalized product

About CHANGING.AI

Founded in March 2018, CHANGING.AI uses as-a-service AI and deep learning to enable online and offline retailers to deliver real-time, personalized recommendations to consumers.

Industries: Technology

Location: Taiwan

Cloud Storage

(<https://cloud.google.com/storage/>), BigQuery.

service recommen-
to consumers.

Google Cloud results

- Plans to deliver deep-learning-based, personalized recommendation services to 20-30 businesses worldwide in 2019
- Pays only for infrastructure and data services used
- Supports daily calculations to understand latest site product information and purchase history
- Enables real-time understanding of consumer purchase intent and combines with lessons from daily calculations

Pla
per
rec
ser
30
20

Google Cloud (<https://cloud.google.com/>)

Operations

Cloud Identity (<https://cloud.google.com/identity/>)

Monitoring

Cloud Dataproc

(<https://cloud.google.com/dataproc/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

(<https://cloud.google.com/container-registry/>)

Cloud Storage (<https://cloud.google.com/storage/>)

(<https://cloud.google.com/load-balancing/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Work

Operations

(<https://cloud.google.com/products/operations>)

Cloud

Monitoring (<https://cloud.google.com/monitoring/>)

Cloud Logging

Logging (<https://cloud.google.com/logging/>)

Cloud

Trace (<https://cloud.google.com/trace/>)

(<https://workspace.google.com/products/gmail/>)

Container Registry

(<https://cloud.google.com/container-registry/>)

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

TensorFlow (<https://www.tensorflow.org/>)

Contact us (/contact)

Google Workspace

(<https://workspace.google.com/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

Google Drive

(<https://workspace.google.com/products/drive/>)

Gmail

(<https://workspace.google.com/products/gmail/>)

to deliver
personalized
recommendations

- Enables rapid
deployment of
applications
through
containerization

Founded in early 2018, Taiwan-headquartered CHANGING.AI (<https://www.changing.ai/>) aims to transform large businesses – particularly retailers – using a-a-service AI and deep learning products.

Headed by the former business and technology leaders of a leading smartphone manufacturer, CHANGING.AI has recruited a skilled team comprising 15 software engineers and the same number of data scientists. The business focuses on developing customer-centric products that meet customer needs.

The business started life focusing on the retail sector – online and offline. "We believe we can help retailers modernize by gaining a better understanding of their customers, and consequently what each customer prefers," says Chia-Lin Chang, Founder and Chief Executive Officer, CHANGING.AI. "This enables retailers to recommend the products and services that best suit the needs of each individual customer."

"Google Cloud
was stable,
highly scalable,
and simple to
operate, so our
decision was
easy."

—Chia-Lin Chang,
Founder and Chief
Executive Officer,
CHANGING.AI

Personalized recommendation powered by deep learning

Deep learning using neural networks – modeled on the human brain and nervous system – drives the value proposition of CHANGING.AI. The business uses deep-learning models to analyze large datasets from historical data and to produce real-time product recommendations.

CHANGING.AI neural networks capture real-time purchase intentions from data about consumers' behavior and use continuous learning to improve the accuracy and relevance of their outputs.

"Traditional machine learning-powered recommendation engines typically rely on models that direct consumers to frequently-viewed items they would be likely to purchase anyway," explains Chang. "However, CHANGING.AI combines a range of algorithms to discover items that are viewed less frequently but incorporate features similar to more popular items. This allows us to recommend long tail products – large volumes of products sold in small numbers – which are typically more profitable to retailers."

The business operates a deep-learning-based recommendation engine primarily serving two types of retailers: large-scale online retailers and retailers with sizeable offline traffic transaction data, plus significant online size and sales.

"In Taiwan, we work with online retailers with monthly traffic of between 3 million and 50 million user sessions, while we offer retailers with an offline and online presence the ability to generate meaningful insights from both sources of data, as well as the combined data set," says Chang.

"Some large ecommerce retailers see an immediate increase in revenue when they deliver personalized, targeted recommendations", he adds, "while others see a steady but still sizeable improvement in revenue and customer experience."

"Google
Kubernetes
Engine is our
container
orchestration
platform and
enables us to
easily develop
applications,
pack them into
containers, and
deploy them
with a single
click."

—Sbin Lin, Chief
Technology Officer,
CHANGING.AI

Cloud agile, scalable, and cost effective

CHANGING.AI needed a platform to run its products and quickly identified the agility, scalability, and cost effectiveness of the cloud as the best fit for its needs. The business' leaders had worked closely

with Google's technical team while at their previous employer and established a deep mutual trust.

"Google Cloud (<https://cloud.google.com/>) was stable, highly scalable, and simple to operate, so our decision was easy," says Chang.

The business deployed Google Cloud using internal resources, with Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>) as the cornerstone of its architecture. "Google Kubernetes Engine is our container orchestration platform and enables us to easily develop applications, pack them into containers, and deploy them with a single click," explains Sbin Lin, Chief Technology Officer, CHANGING.AI. "We use Kubernetes for everything, including all our recommendation services and applications." Using Google Kubernetes Engine to manage hundreds of containerized applications delivers significant improvements in developer productivity, resource efficiency, and operational automation.

The business uses Container Registry (<https://cloud.google.com/container-registry/>) to manage, control access to, and perform vulnerability analyses of its Docker containers, and Cloud Dataproc (<https://cloud.google.com/dataproc/>) to run clusters for data processing and analytics.

CHANGING.AI also uses BigQuery (<https://cloud.google.com/bigquery/>) and is preparing to scale its analytics data warehouse as new clients come on board and the volume of streaming data requiring analysis increases.

The TensorFlow (<https://www.tensorflow.org/>) open source software library and NVIDIA Tesla V100 GPUs train and serve the CHANGING.AI deep learning model. Cloud Storage (<https://cloud.google.com/storage/>) provides file storage services, while Cloud Load Balancing (<https://cloud.google.com/load-balancing/>) manages large volumes of traffic from multiple locations. Stackdriver Monitoring (<https://cloud.google.com/monitoring/>), Stackdriver Logging (<https://cloud.google.com/logging/>) and Stackdriver Trace (<https://cloud.google.com/trace/>) enable CHANGING.AI to view and track application performance.

Cloud Identity (<https://cloud.google.com/identity/>) enables CHANGING.AI to manage access to third-party tools used to manage its products.

"With Google
Cloud, we can
meet our
business
requirement of
completing
training models
within one day to
enable large-

scale
ecommerce
providers to
deliver real-time,
personalized
product
recommendations
to their users."

*—Sbin Lin, Chief
Technology Officer,
CHANGING.AI*

CHANGING.AI is currently exploring the use of Google Cloud services such as Cloud SQL (<https://cloud.google.com/sql/>) for its databases, Cloud Dataflow (<https://cloud.google.com/dataflow/>) for batch and real-time data processing, Cloud Composer (<https://cloud.google.com/composer/>) for workflow orchestration and Looker Studio (<https://lookerstudio.google.com/overview>) for visualization.

Google supports expansion

The competitive pricing of Google Cloud and its pay-as-you-use model enable CHANGING.AI to realize its expansion plans without budgeting large amounts on servers, storage, networking, and other systems.

The business also takes advantage of Google's constant development and regular release of new cloud services to improve its client offerings.

Google Workspace delivering productivity

Lin also cites the ease of managing identity and credentials across Google Cloud services and Google Workspace (<https://workspace.google.com/>), which is deployed across the business to enable staff to collaborate and be productive. Cloud-based intelligent applications such as Sheets (<https://workspace.google.com/products/sheets/>), Drive (<https://workspace.google.com/products/drive/>), and Gmail (<https://workspace.google.com/products/gmail/>) provide business-grade security and easy administration. "Google Workspace is extremely convenient because all services are fully integrated," says Lin.

"With Google Cloud, we can meet our business requirement of completing training models within one day to enable large-scale ecommerce providers to deliver real-time, personalized product recommendations to their users," explains Lin.

Google Cloud is now established as the infrastructure and data platform for CHANGING.AI and the business is well positioned to add new customers and grow into new markets. "In 2019, we are likely to have 20 to 30 big names in the Asia-Pacific, the United States, and Europe," says Chang. "We are in discussions with the Google team as to how best to expand beyond Taiwan and deliver the

light customization needed to meet our customers' specific needs."

**Learn
More**

Reach out to
our team to
see how
Google Cloud
can help your

business. 